ADIL: Adaptive Dual Imitation Learning for Real-Time Object Detection in Remote Sensing Images

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***Abstract*—YOLO, as a representative real-time object detection model, is widely used in remote sensing tasks. However, due to its general-purpose design, it struggles to detect small and dim objects in remote sensing scenarios. While recent improvements to YOLO have focused on improving accuracy, they often overlook computational costs, which limits their deployment on resource-constrained edge devices. To address these challenges, we propose adaptive dual imitation learning (ADIL), a novel strategy designed to enhance the feature representation of small and dim objects in remote sensing scenes. ADIL equips the YOLO detector with a feature imitation (FI) branch, which is discarded during inference to maintain efficiency. Specifically, ADIL combines self-imitation and VLM-imitation learning strategies, each constructing exemplar sets to guide the feature learning process. In self-imitation learning, high-quality samples are dynamically selected based on the network's learning state to build the exemplar sets. In VLM-imitation learning, text-space features from a vision-language model are used to construct the exemplar sets. Additionally, a contrastive learning-based imitation loss is applied to optimize the imitation branch. When integrated into YOLO, ADIL significantly improves detection performance on multiple remote sensing datasets while preserving the original speed of the detector, achieving a better balance between accuracy and efficiency. We hope that this approach provides valuable insights for real-time remote sensing object detection.**

***Index Terms*—Imitation learning, object detection, remote sensing, YOLO.**

I. Introduction

REMOTE sensing images have been widely applied across various tasks, including change detection [1], [2], visual question answering [3], clustering [4], [5], land cover classification [6], and object detection [7], [8], [9], [10]. Among these, remote sensing image object detection (RSOD) plays a critical role in several fields, such as disaster monitoring, urban planning, environmental protection, and military surveillance. Due to its broad range of applications, RSOD has become an important research topic in the field of computer vision [7], [11], [12], [13]. In many RSOD applications, real-time processing is often a key requirement to ensure timely decision-making and response. As a result, one-stage detection models, which balance speed and accuracy, are the most commonly used in RSOD. Among these, YOLO [8], [9], [14], [15], [16], [17], [18], [19], [20], [21], [22] is one of the most representative and widely adopted models.

Compared to general object detection scenarios, remote sensing images present unique challenges due to their distinct characteristics. These include small-object sizes, weak visible information, cluttered backgrounds, occlusions, and large variations in object scales and orientations. Most existing object detectors are designed for general scenarios and struggle to handle the specific challenges of remote sensing data, which results in suboptimal performance.

Several methods have been proposed to enhance the accuracy of RSOD from the perspective of feature enhancement, such as [9], [23], and [24]. Although these methods have shown promising performance, they bring extra comuptation burden which restricts their deployment on edge devices and real-time systems.

Recently, imitation learning has been utilized to bridge the representation gap between small objects and large ones providing a more efficient way to improve the performance of hard examples. However, most of them are designed for two-stage detection models in general scenarios and they cannot be directly applied to YOLO in remote sensing images. The reasons are as follows. First, YOLO [14], [15], [16], [17], [18], [19], [20], [21], [25] is the anchor-based one-stage model and its object feature representation mechanism significantly differs from proposal-based two-stage detection models [26], making existing methods unsuitable for direct application in remote sensing scenarios. Second, compared to general scenarios, remote sensing data have a much limited size and the feature of large objects may not provide good guidance for the small object to learn. Finally, current methods rely on IoU to distinguish high- and low-quality samples, but high IoU does not necessarily indicate high quality. True high-quality samples should perform well in both classification and localization. In a word, how to apply imitation learning to YOLO in remote sensing images is still underexplored.

Moreover, the HQ-based self-imitation paradigm has its limitations in remote sensing. Given that remote sensing often involves small, dim, and weak objects, the number of high-quality samples is relatively low, and feature representations are unstable—especially in the early training process. These factors provide the network with limited and ambiguous supervisory signals, which constrain training efficiency.

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Besides, self-imitating learning captures predominantly in-domain knowledge, which may lead to overfitting and prevent the model from learning more generalized knowledge.

To this end, we fill this gap by proposing an adaptive dual imitation learning (ADIL) strategy tailored to the characteristics of remote sensing scenarios. This strategy includes self-imitation and VLM-imitation. In self-imitation learning, the features of high-quality samples are used to form an exemplar set, which guides the learning of low-quality samples. To better distinguish between high- and low-quality samples, we propose a quality metric that dynamically evaluates sample quality based on the outputs of the network's classification and regression branches. What is more, considering the potential inefficiency of self-imitation caused by limited data size in the remote sensing area, we creatively introduce a VLM-imitation strategy. This approach leverages the powerful feature representation capabilities of vision-language models (VLMs) to guide the learning of hard samples. Since VLMs align image and text feature spaces well, we can only use the text embeddings of category names to construct the exemplar set, which is static during the whole training process, offering consistent category priors and more reliable and general supervision when HQ visuals are insufficient. In a word, self-imitation learning provides the foundational in-domain knowledge, while VLM-imitation learning supplies general-domain knowledge. The two complement each other, jointly delivering reliable and robust supervisory signals.

To implement our learning strategy, we equip the original detection head with an additional imitation branch, which is optimized using supervised contrastive learning between low-quality samples and the representative and discriminative exemplar set. This branch can be discarded during inference, maintaining the fast speed of the original detector. Experiments on various-sized YOLOv8/YOLOv10 architectures and multiple datasets demonstrate that our method achieves significantly better performance than the baseline, validating its effectiveness.

In summary, our main contributions are as follows.

1) We are the first to apply imitation learning in one-stage remote sensing object detection.
2) We propose a novel ADIL strategy that combines the network's own features with VLM features to enhance the feature learning of challenging objects.
3) We validate the effectiveness of our method on multiple datasets, achieving a better balance between accuracy and speed.

II. Related Works

A. Feature Enhancement for Object Detection

In remote sensing and aerial images, small and dim objects are commonly present. Existing methods for addressing RSOD [7], [8], [9], [10], [11], [13], [21], [27], [28], [29] primarily focus on improving the network's representation ability with extra feature enhancement modules. To enhance the utilization of multiscale feature fusion, Zhang et al. [8] proposed FFCA-YOLO with three lightweight modules: feature enhancement module, feature fusion module, and spatial context aware module to enhance local perception, multiscale feature fusion, and global correlations, improving small-object detection and suppressing background noise with minimal complexity. Xiao et al. [30] introduced a new feature pyramid network, which primarily consists of context augmentation module and feature refinement module, enhancing contextual features and refining features. They also proposed a novel data augmentation method, copy-paste, to increase the contribution of small objects to the loss during training, ensuring a more balanced training process. To better leverage contextual information, Li et al. [13] proposed LSKNet, which employs a large selective kernel to dynamically adjust and obtain a larger spatial receptive field, enabling better modeling of long-range context in remote sensing scenes. Shi et al. [31] proposed a progressive class-aware instance enhancement strategy for aircraft detection, strengthening instance-level cues to alleviate background interference and small-occupancy targets. Meanwhile, Hu et al. [32] explored optical–SAR complementarity and introduced an unpaired Opt2SAR translation with complementarity-aware feature fusion to enhance aircraft detection without paired registrations. Shi et al. [33] introduced similarity distance to evaluate the similarity between bounding boxes, considering both positional and shape similarities. Zhang et al. [34] proposed SO-DETR, a detection transformer-based model specifically designed for small-object detection. By integrating a dual-domain hybrid encoder and an improved query selection mechanism, SO-DETR effectively fuses spatial and frequency-domain features to capture high-resolution details more accurately and enhance query allocation efficiency. Although these methods have achieved promising performance, they introduce additional parameters and hardware-unfriendly operators, making them difficult to apply to RSOD with high real-time requirements.

B. Imitation Leaning for Object Detection

The main challenges in detecting tiny and dim objects include their minimal pixel presence, which increases susceptibility to background noise, and their tight arrangement, leading to feature interference and the insufficient availability of discriminative features. These issues have driven many researchers to explore how to utilize discriminative and high-quality semantic features to enhance detection performance for targets with unclear semantic features. Wu et al. [35] designed a mimic loss based on Faster R-CNN [26] and proposed SML, which primarily forces the feature representation of small-scale pedestrians to approach that of large-scale pedestrians. Kim et al. [36] proposed a pedestrian detection framework based on Faster R-CNN. This framework constructs an offline LPR and aligns the features of small-scale pedestrians with the large-scale pedestrian features stored in the memory. In addition, Yuan et al. [37] proposed CFINet by adopting CRPN and FI to ensure the acquisition of higher quality small-scale targets and enhance the feature representation of small-scale targets. Although these methods have achieved good performance in detecting challenging small objects, they are designed for two-stage detection algorithms in nonremote sensing scenarios. Due to the significant differences in network structures between YOLO and two-stage detection models, as

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well as the disparities in data scale and scene characteristics between remote sensing and general scenarios, these methods cannot be directly and effectively applied to YOLO-based remote sensing detection tasks. In this article, our goal is to bridge this gap by proposing an effective and efficient imitation learning strategy for YOLO-based RSOD.

C. Object Detection Based on Contrastive Learning

Contrastive learning [23], [38] is one of the key methods in the field of self-supervised learning. The main idea is to encourage similar instances to be mapped closer together in the learned embedding space, while pushing dissimilar instances further apart. Some research has extended the concept of contrastive learning to the detection domain. Xie et al. [24] proposed DetCo, enhancing MoCov2 [39] with multistage and cross-scale contrastive learning for improved detection performance. Kumar et al. [40] integrated interimage and intraimage contrastive techniques to capture location information, simplifying object detection labeling. Wang et al. [41] designed a comprehensive few-shot object detection model structure based on contrastive learning, addressing issues such as misclassification of new instances and inaccurate object localization. Biswas and Tešić [42] used contrastive learning to align local and global features obtained from the backbone, mitigating the impact of previously unseen object classes on detection performance, and enhancing the domain generalization capability of detection algorithms. Tu et al. [43] introduced contrastive learning into YOLOv5 by constructing contrastive losses through random cropping of target and background image regions from image samples. Despite the growing attention to contrastive learning recently [23], [38], [44], [45], the potential of using contrastive learning for imitation learning to better enhance the detection of weak and dim objects with one-stage detectors has yet to be explored.

III. Methodology

The core idea of imitation learning is to encourage training samples in the current batch to mimic representative exemplar features, thereby enhancing the network's feature learning capabilities. As illustrated in Fig. 1, we integrate an imitation branch into the one-stage YOLOv10 [19] detector to implement our ADIL framework, which consists of self-imitation learning and VLM-imitation learning. Self-imitation learning, similar to existing methods, leverages the detector itself to extract high-quality sample features for constructing exemplars. In contrast, VLM-imitation learning utilizes exemplars derived from a text feature space aligned with the visual domain, such as CLIP's [46] textual embeddings. These two approaches work in tandem, offering guidance that combines domain-specific information with robust and semantically rich feature representations.

We first describe the process of selecting high-quality samples using the proposed quality score. Next, we explain how high-quality samples and vision-language models are used to construct exemplar features for self-imitation and VLM-imitation, respectively. Finally, we detail the optimization and training procedure.

A. Definition of High-Quality Samples

Existing methods mostly use object size or the IoU of predicted bounding boxes as criteria to classify sample quality. We argue that such criteria are unsuitable for detection tasks because they do not consider the network's learning state and are inconsistent with current mainstream label assignment strategies. Object detection involves both classification and regression, so an appropriate criterion should take into account both classification scores and regression quality. Based on this, we propose a quality score (QS) to select low-quality samples.

Suppose that the outputs of the regression branch (RI) are $O_{\text{reg}} \in \mathbb{R}^{B\times N\times 4}$ and the outputs of the classification branch (CI) are $O_{\text{cls}} \in \mathbb{R}^{B\times N\times C}$, where B denotes the batch size, N is the number of anchor points per image, and C represents the number of object categories. After label assignment [18], there are N_{pos} positive samples remained. By jointly considering the classification confidence and localization accuracy, we assign each positive sample a quality score t as

$$f\left(s^{\alpha}\cdot\mu^{\beta}\right) \tag{1}$$

where s and μ denote the classification score and the intersection-over-union (IoU) between the predicted box and the ground truth, respectively. The exponents α and β are hyperparameters that control the relative importance of classification and localization quality in the alignment score. The function $f(\cdot)$ denotes a scale normalization operation to ensure training stability.

A higher value of t indicates that the anchor is both more confident and more accurate in localization. By applying an appropriate threshold t_{hq}, we can select proper high-quality samples which can be used to form the exemplar set in self-imitation.

B. Extraction of Exemplar Features

As shown in Fig. 1, the feature imitation head (FIH) in the imitation branch (IB) utilizes several convolution layers to project the output features from the neck to the embedding space with dimension d. The imitation process is performed in such embedding space. The parameters of FIH have a similar structure with the classification branch and are shared across different feature levels. The output of ID for each image can be denoted as $F_{ib} = \{f_{ib}^{i}\}_{i=1}^{N}$, where N denotes the total number of anchors in the image. $f_{ib}^{i} \in \mathbb{R}^{d}$ denotes the ith anchor embedding with dimension d.

1) Self-Imitation Learning: In self-imitation learning, we maintain a dynamic memory bank of class exemplars (MBCE). The dynamic class exemplars for the dataset are denoted as $F_{mb} = \{f_{mb}^{i}\}_{i=1}^{C}$, where C is the number of object categories, and each exemplar $f_{mb}^{c} \in \mathbb{R}^{d}$ corresponds to the cth class, with d being the feature dimension. Initially, all exemplars are initialized to zero. The MBCE operates as an online updatable memory module designed to retain class-level knowledge and guide instance-level feature learning.

High-quality positive (HQP) samples—identified as those with the quality score t exceeding a threshold t_{hq}—are used to update the memory bank. Each HQP sample is represented by an object embedding $f_{ib_hq} \in \mathbb{R}^{d}$ received from the imitation

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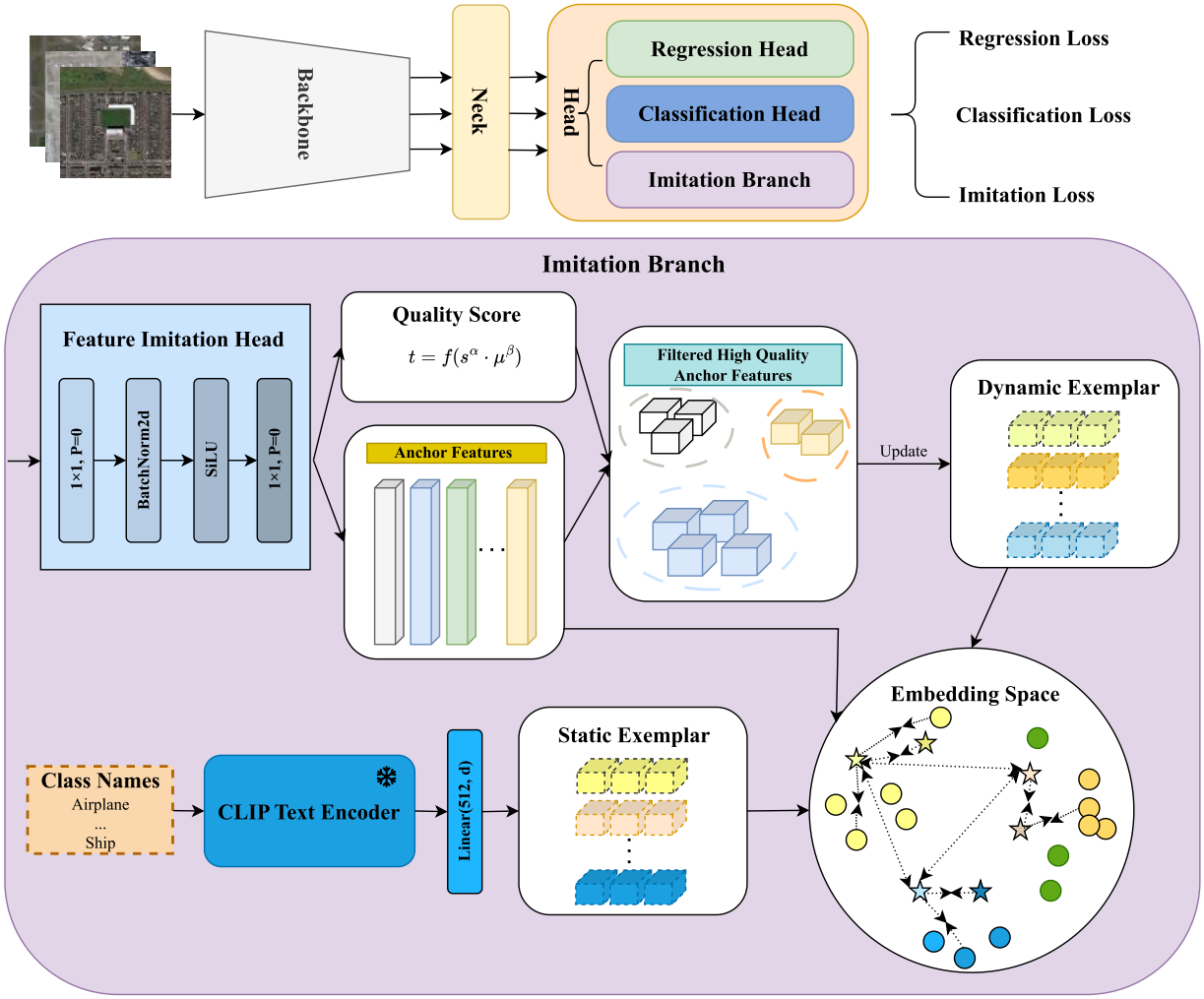


Fig. 1. Overall framework of ADIL is designed to enhance the representation of small and dim anchors. In the imitation branch, anchor features are guided to align with exemplar features of the same category in the embedding space while being repelled from those of other categories. This branch comprises two components: self-imitation, where dynamic exemplars are generated from high-quality anchors in previous iterations using a moving average approach, and VLM-imitation, where exemplars are derived from the CLIP text encoders. The optimization of the imitation branch is driven by our proposed imitative learning loss.

branch and its associated class label $l_{ib_hq} \in \{1, \ldots, C\}$. The memory update follows a momentum-based scheme. For each class c, the current class center is computed from all HQP samples in the batch and used to update the corresponding exemplar in the memory bank. The update process is formulated as follows:

$$f^c_{\text{cur}} = \frac{1}{\sum_{j=1}^{H} \mathbb{I}\left(l^j_{ib_hq} = c\right)} \sum_{j=1}^{H} f^j_{ib_hq} \times \mathbb{I}\left(l^j_{ib_hq} = c\right)$$
$$f^c_{mb} = m \times f^c_{mb'} + (1 - m) \times f^c_{\text{cur}} \tag{2}$$

where f^c_{cur} denotes the current batch-wise center for class c, f^c_{mb} is the updated class exemplar, H is the number of HQP samples in the batch, and $m \in [0, 1]$ is a momentum coefficient controlling the update rate. The indicator function $\mathbb{I}(l^j_{ib_hq} = c)$ returns 1 if the jth HQP sample belongs to class c and 0 otherwise.

This adaptive updating mechanism enables the class exemplars to evolve dynamically throughout training, improving their representation capacity and contributing to more discriminative feature learning under the feature imitation paradigm.

2) VLM-Imitation Learning: In VLM-imitation learning, we represent each category's exemplar features using the corresponding CLIP textual features. Specifically, for each category, its name (e.g., "car" or "airplane") is input into the CLIP text encoder to obtain the normalized textual feature embedding. These embeddings are stored in a memory bank and serve as the exemplar features throughout training. The exemplar features are denoted as $F_{clip} = \{f^i_{clip}\}_{i=1}^{C}$, where C is the number of classes. The ith text embedding f^i_{clip} is calculated as $f^i_{clip} = E_{text}(P_i)$, where E denotes the text encoder of CLIP and P_i indicates the prompt of ith category (e.g., "a photo of car").

By leveraging CLIP's pretraining on large-scale image-text pairs, these textual exemplars provide rich semantic information, which guides low-quality samples in learning more discriminative features. The class-wise exemplar features in

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VLM-imitation are computed in advance and kept static during the whole training process.

C. Imitative Contrastive Loss

1) Instance-Level Loss: The optimization objective of imitation learning is conceptually straightforward. Leveraging supervised contrastive learning, our ADIL measures the similarity between the positive anchor embeddings defined by the label assignment and the updated class exemplars (F_{mb} and F_{clip}). The objective is to pull these sample features closer to the exemplars of their corresponding categories while simultaneously pushing them away from those of other categories.

To this end, we define an instance-level imitation loss as follows:

$$\ell_i^a = -\log \frac{\exp\left(\frac{sim\left(\boldsymbol{f}_{ib}^i, \boldsymbol{f}_{mb}^j\right)}{\tau}\right) + \exp\left(\frac{sim\left(\boldsymbol{f}_{ib}^i, \boldsymbol{f}_{clip}^j\right)}{\tau}\right)}{\sum_{k=1}^{C}\left[\exp\left(\frac{sim\left(\boldsymbol{f}_{ib}^i, \boldsymbol{f}_{mb}^k\right)}{\tau}\right) + \exp\left(\frac{sim\left(\boldsymbol{f}_{ib}^i, \boldsymbol{f}_{clip}^k\right)}{\tau}\right)\right]} \tag{3}$$

$$\mathcal{L}_{\text{IIL}} = -\frac{1}{M}\sum_{i=1}^{M}\ell_i^a \tag{4}$$

where $\text{sim}(\cdot,\cdot)$ denotes the cosine similarity function and τ is a temperature parameter controlling the concentration of the similarity distribution, which requires careful tuning. Here, C is the number of classes and M is the number of positive samples. The embedding vector $\boldsymbol{f}_{ib}^i$ corresponds to the embedding of the ith positive sample whose GT class label is j. $\boldsymbol{f}_{mb}^j$ and $\boldsymbol{f}_{clip}^j$ denote the dynamic and static class exemplar embeddings of class j, respectively. This loss encourages sample features to align with the representations of both foundation model and domain-specific model, thus improving the discriminability and generalization of learned features.

2) Class-Level Loss: Apart from the instance-level loss, we introduce a *class-level contrastive loss* to promote both consistency and diversity in class-level representations. The loss is defined as

$$\hat{\ell}_i^a = -log\frac{\exp\left(\frac{sim\left(\boldsymbol{f}_{\text{mb}}^i, \boldsymbol{f}_{\text{mb}'}^i\right)}{\tau}\right) + \exp\left(\frac{sim\left(\boldsymbol{f}_{\text{mb}}^i, \boldsymbol{f}_{\text{clip}}^i\right)}{\tau}\right]}{\sum_{j=1}^{C}\left[\exp\left(\frac{sim\left(\boldsymbol{f}_{\text{mb}}^i, \boldsymbol{f}_{\text{mb}'}^j\right)}{\tau}\right) + \exp\left(\frac{sim\left(\boldsymbol{f}_{\text{mb}}^i, \boldsymbol{f}_{\text{clip}}^j\right)}{\tau}\right)\right]} \tag{5}$$

$$\mathcal{L}_{\text{CIL}} = -\frac{1}{2C}\sum_{i=1}^{C}\hat{\ell}_i^a \tag{6}$$

where $\text{sim}(\cdot,\cdot)$ denotes the cosine similarity function and τ is a class-level temperature parameter shared with the instance-level loss to ensure consistent scaling of logits. C is the number of object categories. The vector $\boldsymbol{f}_{mb'}^i$ denotes the class exemplar for category i before the update, while $\boldsymbol{f}_{\text{mb}}^i$ and $\boldsymbol{f}_{\text{mb}}^j$ refer to the updated exemplars for categories i and j, respectively. Similarly, $\boldsymbol{f}_{\text{clip}}^i$ and $\boldsymbol{f}_{\text{clip}}^j$ represent the CLIP-encoded semantic features for categories i and j. This loss encourages alignment between the pre- and post-update class exemplars while simultaneously preserving interclass separability.

Algorithm 1 Training Procedure of the Imitation Branch

Input:
A batch $\mathcal{B} = \{x_i\}_{i=1}^{B}$, and labels $\mathcal{Y} = \left\{\left\{(y_i^j, \text{bbox}_i^j)\right\}_{j=1}^{G_i}\right\}_{i=1}^{B}$
Exemplars in VLM-imitation $F_{\text{clip}} \in \mathbb{R}^{C\times d}$
Exemplars in self-imitation before updating $F_{\text{mb}'} \in \mathbb{R}^{C\times d}$
Threshold t_{hq}; Momentum m; Weight coefficient w_{IL}

Output: Updated memory bank F_{mb}

Extract outputs: $O_{\text{cls}}, O_{\text{reg}}, F_{\text{ib}}$
Compute quality scores via Eq. (1)
Select high-quality samples $F_{\text{ib_hq}}$
for *each class* $c = 1$ *to* C **do**
  Compute class center: $f_{\text{cur}}^c \leftarrow \text{mean}(F_{\text{ib_hq}}^c)$
  Update memory bank: $f_{\text{mb}}^c \leftarrow m \cdot f_{\text{mb}'}^c + (1-m)\cdot f_{\text{cur}}^c$
Compute $\mathcal{L}_{\text{IL}} \leftarrow \mathcal{L}_{\text{IIL}} + \mathcal{L}_{\text{CIL}}$
Compute total loss $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + w_{\text{IL}}\cdot\mathcal{L}_{\text{IL}}$
Backpropagate and update model

3) Total Loss: The total loss function is defined as

$$\mathcal{L}_{\text{overall}} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} + w_{\text{IL}}\left(\mathcal{L}_{\text{IIL}} + \mathcal{L}_{\text{CIL}}\right) \tag{7}$$

where w_{IL} is a weighting coefficient that controls the contribution of the imitative contrastive losses to the overall optimization objective. The terms $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{reg}}$ denote the standard classification and regression losses used in the original detection framework.

Notably, the imitation learning process is employed exclusively during the training phase and is removed during inference. This design ensures that the proposed method introduces no additional computational overhead at test time, thereby preserving the inference speed and runtime efficiency of the baseline detector. Algorithm 1 outlines the details of the imitation learning process.

IV. Experiments

In this section, we first introduce the datasets and evaluation protocols employed in our experiments. Subsequently, we provide a detailed description of the implementation details. Finally, we present comprehensive comparisons with state-of-the-art methods and conduct extensive ablation studies to thoroughly evaluate the effectiveness of the proposed approach.

A. Datasets and Evaluation Metrics

1) Datasets: To evaluate the effectiveness and universality of the proposed method, we adopt **YOLOv8** [17] and **YOLOv10** [19] as the baseline detector. By integrating the proposed ADIL strategy into its training pipeline, we develop an enhanced variant, referred to as **YOLOv8-ADIL** and **YOLOv10-ADIL**. Extensive experiments are conducted on

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three widely used remote sensing and aerial image benchmarks: **DIOR** [18], **AI-TOD-v2** [12], and **VisDrone2019-Det** [20]. Detailed information of datasets is shown in Table I.

a) DIOR: The DIOR dataset comprises 23 463 remote sensing images, each with a spatial resolution of 800 × 800 pixels. It contains 190 288 annotated instances in the form of horizontal bounding boxes, covering 20 object categories, including *basketball court*, *vehicle*, *tennis court*, *airplane*, *storage tank*, *bridge*, *windmill*, *baseball field*, *ground track field*, *stadium*, *expressway toll station*, *dam*, *golf field*, *ship*, *overpass*, *train station*, *airport*, *chimney*, *expressway service area*, and *harbor*. The dataset is characterized by significant variations in object scale, orientation, and spatial density. The average absolute object size is 65.7 pixels.

b) AI-TOD-v2: The Aerial Image Target Detection-Version 2 (AI-TOD-v2) dataset is a large-scale benchmark tailored for complex aerial object detection scenarios. Compared to its predecessor AI-TOD, this version provides broader scene diversity, higher annotation granularity, and increased dataset scale. It comprises 28 036 high-resolution aerial images of size 800 × 800 pixels, with 752 745 annotated objects spanning eight categories: *airplane*, *bridge*, *storage tank*, *ship*, *swimming pool*, *vehicle*, *person*, and *windmill*. With an average object size of just 12.7 pixels, AI-TOD-v2 poses a substantial challenge for small-object detection.

c) VisDrone2019-Det: The VisDrone2019-Det dataset consists of 8629 aerial images with a resolution of 640 × 640 pixels and includes 10 209 annotated instances. It covers ten object categories: *pedestrian*, *people*, *bicycle*, *car*, *van*, *truck*, *tricycle*, *awning-tricycle*, *bus*, and *motor*. This dataset presents significant challenges due to a combination of dense object distributions, variable scales, and diverse object orientations. The average absolute object size is 35.8 pixels.

2) Evaluation Metrics: To comprehensively evaluate the performance of the proposed method, we adopt the widely recognized mean average precision (mAP) metric, following the standard COCO evaluation protocol. This metric is extensively utilized to objectively assess the effectiveness of object detection and instance segmentation algorithms. Specifically, a detection result is considered a true positive if the predicted bounding box or mask achieves an IoU with the corresponding ground truth greater than a predefined threshold T, and the predicted category label matches the ground truth category.

In this study, we report AP, AP_{50}, AP_{Small}, AP_{Medium}, and AP_{Large} to provide a comprehensive assessment of detection performance. The AP metric represents the average precision calculated over ten IoU thresholds ranging from 0.50 to 0.95 with a step size of 0.05, averaged across all categories. A higher AP score indicates more accurate localization and classification, reflecting superior detection performance. AP_{50} denotes the average precision under a fixed IoU threshold of 0.50. Additionally, AP_{Small}, AP_{Medium}, and AP_{Large} evaluate performance on objects of different scales, providing a more granular analysis of detection effectiveness across varying object sizes.

B. Implementation Details

In the following experiments, for both the DIOR and AI-TOD-v2 datasets, training is conducted on the union of the training and validation sets, while the test set is reserved exclusively for performance evaluation and ablation studies. For the VisDrone2019-Det dataset, training is performed solely on the training set, with the validation set used exclusively for evaluation and ablation purposes.

All experiments are conducted on a machine equipped with four NVIDIA V100 GPUs. During training and validation, the input image resolution is set to 800 × 800 pixels for DIOR and AI-TOD-v2, and 640 × 640 pixels for VisDrone2019-Det.

The dimensionality d of both the class exemplars and the feature embeddings is set to 128. Unless otherwise specified, the high-quality threshold t_{hq} is fixed at 0.7. The exponents α and β used in computing the quality score for each positive sample are set to 0.5 and 6.0, respectively. The momentum coefficient m used for updating class exemplars is fixed at 0.9. The temperature parameter τ, shared by both the instance-level loss $\mathcal{L}_{\text{IIL}}$ and the class-level loss $\mathcal{L}_{\text{CIL}}$, is set to 0.6. Additionally, the loss weight w_{IL} is set to 0.5. Unless otherwise noted, all remaining hyperparameters are aligned with those used in the baseline detection models. The number of training epochs is fixed at 600.

C. Main Results

To validate the effectiveness of our method, we conduct a thorough comparison with existing representative detectors across three datasets. The optimal results are **highlighted in boldface** in Tables II–IV.

As shown in Table II, we evaluate the performance of our proposed YOLOv8-ADIL and YOLOv10-ADIL models against a range of existing object detectors on the **VisDrone2019-Det** dataset. Across all three model scales (S, M, and L), integrating the ADIL module into YOLOv8/YOLOv10 consistently improves both AP and AP_{50} metrics. Specifically, YOLOv8-ADIL-S/YOLOv10-ADIL-S, YOLOv8-ADIL-M/YOLOv10-ADIL-M, and YOLOv8-ADIL-L/YOLOv10-ADIL-L achieve absolute AP gains of +1.3%/+1.7%, +1.9%/+1.9%, and +2.0%/+2.2%, respectively, over their baseline counterparts. Corresponding improvements in $\mathbf{AP_{50}}$ are +1.6%/+2.5%, +2.8%/+2.9%, and +2.7%/+3.1%. These results clearly demonstrate the effectiveness of the ADIL module in enhancing detection performance. Notably, these performance gains are achieved without introducing any additional computational overhead or parameters during inference, underscoring the efficiency and practicality of the proposed design.

To further validate the effectiveness and generalizability of our approach, we conduct experiments on two additional datasets: **DIOR** and **AI-TOD-v2**. As shown in Table III, the proposed ADIL series consistently outperforms the baseline YOLOv8/YOLOv10 models across all scales. Specifically, YOLOv8-ADIL-S/YOLOv10-ADIL-S, YOLOv8-ADIL-M/YOLOv10-ADIL-M, and YOLOv8-ADIL-L/YOLOv10-ADIL-L achieve AP improvements of +1.4%/+1.1%, +1.2%/+1.5%, and +1.8%/+1.5%,

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TABLE I
DATASETS DESCRIPTION. "-/-" INDICATES THE NUMBER OF IMAGES AND INSTANCES IN THE DATASET, RESPECTIVELY

Dataset	Classes	Image Size	Train	Val	Test	All
DIOR [11]	20	800	5862 / 32634	5863 / 35439	11738 / 124445	23463 / 190288
AI-TOD-v2 [12]	8	800	11214 / 301534	2804 / 75091	14018 / 376129	28036 / 752754
VisDrone2019-Det [47]	10	640	6471 / 343205	548 / 38759	1610 / 75102	8629 / 457066

TABLE II
EXPERIMENTAL RESULTS ON THE VISDRONE2019-DET

Model	Backbone	Input Size	Params(M)	GFLOPs(G)	AP(%)	AP_{50}(%)
YOLOv8-S [17]	CSPDarknet(C2F)	640×640	11.1	28.5	22.3	37.6
YOLOv9-S [18]	CSPDarknet(GELAN, PGI)	640×640	7.2	26.7	22.7	38.3
YOLOv10-S [19]	CSPDarknet(SCDown)	640×640	7.2	21.6	22.2	37.4
YOLOv11-S [20]	CSPDarknet(C3K2, C2PSA)	640×640	9.4	21.3	23.0	38.7
HIC-YOLOv5 [48]	CSPDarknet	640×640	9.4	31.2	20.8	36.1
YOLOv8-ADIL-S	**CSPDarknet(C2F, FIL)**	640×640	11.1	28.5	23.5(+1.3)	39.2(+1.6)
YOLOv10-ADIL-S	**CSPDarknet(SCDown, FIL)**	640×640	7.2	21.6	**23.9(+1.7)**	**39.9(+2.5)**
YOLOv8-M [17]	CSPDarknet(C2F)	640×640	25.9	78.9	24.6	40.7
YOLOv9-M [18]	CSPDarknet(GELAN, PGI)	640×640	20.1	76.8	25.2	42.0
YOLOv11-M [20]	CSPDarknet(C3K2, C2PSA)	640×640	20.1	68.0	25.5	42.3
RT-DETR-RN18 [49]	ResNet18	640×640	20.0	60.0	**26.7**	**44.6**
YOLOv10-M [19]	CSPDarknet(SCDown)	640×640	15.4	59.1	24.5	40.5
YOLOv8-ADIL-M	**CSPDarknet(C2F, FIL)**	640×640	25.9	78.9	26.5(+1.9)	43.5(+2.8)
YOLOv10-ADIL-M	**CSPDarknet(SCDown, FIL)**	640×640	15.4	59.1	26.4(+1.9)	43.4(+2.9)
YOLOv8-L [17]	CSPDarknet(C2F)	640×640	43.7	165.2	26.1	42.7
YOLOv11-X [20]	CSPDarknet(C3K2, C2PSA)	640×640	56.8	194.5	28.1	45.6
DetectorRS w/RFLA [50]	ResNet50	800×800	123.2	160.0	27.4	45.3
DCFL [51]	ResNet50	1024×1024	36.1	157.8	-	32.1
Faster R-CNN [26]	ResNet50	640×640	41.3	90.4	26.2	42.8
DETR [52]	ResNet50	1333×750	60.0	187.0	24.1	40.1
Deformable DETR [53]	ResNet50	1333×800	40.0	173.0	27.1	42.2
Sparse DETR [54]	ResNet50	1333×800	40.9	121.0	27.3	42.5
RT-DETR-RN50 [49]	ResNet50	640×640	42.0	130.5	28.4	**47.0**
YOLOv10-L [19]	CSPDarknet(SCDown)	640×640	24.4	120.3	26.3	43.1
YOLOv8-ADIL-L	**CSPDarknet(C2F, FIL)**	640×640	43.7	165.2	28.1(+2.0)	45.4(+2.7)
YOLOv10-ADIL-L	**CSPDarknet(SCDown, FIL)**	640×640	24.4	120.3	**28.5(+2.2)**	46.2(+3.1)

respectively, while the corresponding AP_{50} gains are +1.8%/+2.5%, +2.2%/+2.4%, and +1.6%/+2.0%. These results indicate that the proposed FIL module significantly enhances detection accuracy even on complex and large-scale remote sensing datasets such as DIOR.

Similarly, as presented in Table IV, the YOLOv8-ADIL/YOLOv10-ADIL models also deliver superior performance on the **AI-TOD-v2** dataset, which is known for its extreme object scale variation and dense distributions. YOLOv8-ADIL-S/YOLOv10-ADIL-S, YOLOv8-ADIL-M/YOLOv10-ADIL-M, and YOLOv8-ADIL-L/YOLOv10-ADIL-L achieve AP gains of +1.3%/+1.6%, +2.2%/+1.8%, and +1.8%/+1.5%, and AP_{50} gains of +1.8%/+2.3%, +2.3%/+2.1%, and +2.2%/+1.8%, respectively, compared to their YOLOv8/YOLOv10 counterparts.

The comprehensive comparative results clearly demonstrate that our ADIL strategy significantly enhances the original YOLOv8/YOLOv10 detectors' ability to learn features for detecting weak and dim targets. This enhancement not only improves detection performance but also maintains the original parameter size and computational efficiency during inference.

D. Ablation Study

In this section, we systematically evaluate the influence of key components in ADIL on detection performance and training robustness. We first investigate different exemplar generation strategies, including self-imitation exemplars, VLM-based exemplars, and their dual-exemplar coupling, to verify the necessity and complementarity of each exemplar source. We then analyze the high-quality sample selection metric, with a particular focus on the exponent β that adjusts the relative emphasis of localization reliability (IoU) versus classification confidence. Next, we study the effectiveness of the instance-level imitation loss $\mathcal{L}_{IIL}$, the class-level imitation loss $\mathcal{L}_{CIL}$, and their combination, in order to reveal the contribution of fine-grained instance matching and semantic class alignment. Finally, we report sensitivity results of the

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TABLE III
EXPERIMENTAL RESULTS ON THE DIOR

Model	Backbone	Input Size	Params(M)	GFLOPs(G)	AP(%)	AP_{50}(%)
YOLOv5-S [14]	CSPDarknet(C3)	800×800	7.01	16.02	47.8	78.9
YOLOv8-S [17]	CSPDarknet(C2F)	800×800	11.2	44.7	58.8	80.1
YOLOv10-S [19]	CSPDarknet(SCDown)	800×800	7.2	33.7	59.6	80.3
YOLOv8-ADIL-S	**CSPDarknet(C2F, FIL)**	800×800	11.2	44.7	60.2(+1.4)	81.9(+1.8)
YOLOv10-ADIL-S	**CSPDarknet(SCDown, FIL)**	800×800	7.2	33.7	**60.7(+1.1)**	**82.8(+2.5)**
YOLOv8-M [17]	CSPDarknet(C2F)	800×800	25.9	123.3	61.7	81.5
YOLOv10-M [19]	CSPDarknet(SCDown)	800×800	15.4	92.4	60.8	81.7
YOLOv8-ADIL-M	**CSPDarknet(C2F, FIL)**	800×800	25.9	123.3	**62.9(+1.2)**	83.7(+2.2)
YOLOv10-ADIL-M	**CSPDarknet(SCDown, FIL)**	800×800	15.4	92.4	62.3(+1.5)	**84.1(+2.4)**
ATSS [55]	ResNet50	800×800	32.11	80.48	50.4	78.9
RetinaNet [56]	ResNet50	-	36.33	81.57	44.0	68.3
FCOS [57]	ResNet50	-	32.11	78.58	46.2	72.8
CenterNet [58]	ResNet50	-	32.11	78.56	47.3	73.8
Faster R-CNN [26]	ResNet50	-	41.3	90.4	45.1	64.9
Cascade-RCNN [59]	ResNet50	-	62.15	121.86	46.2	65.0
Mask-RCNN [60]	ResNet50	-	43.97	145.41	45.3	63.1
YOLOv8-L [17]	CSPDarknet(C2F)	800×800	43.7	258.1	61.9	82.6
DETR [52]	ResNet50	-	60.0	187.08	38.7	69.1
RT-DETR-RN50 [49]	ResNet50	-	42.0	130.5	48.3	75.1
Deformable-DETR [53]	ResNet50	-	40.0	173.0	50.9	82.6
YOLOv10-L [19]	CSPDarknet(SCDown)	800×800	24.4	188.0	61.0	82.2
YOLOv8-ADIL-L	**CSPDarknet(C2F, FIL)**	800×800	43.7	258.1	**63.7(+1.8)**	**84.2(+1.6)**
YOLOv10-ADIL-L	**CSPDarknet(SCDown, FIL)**	800×800	24.4	188.0	62.5(+1.5)	**84.2(+2.0)**

TABLE IV
EXPERIMENTAL RESULTS ON THE AI-TOD-V2

Model	Backbone	Input Size	Params(M)	GFLOPs(G)	AP	AP_{50}
YOLOv8-S [17]	CSPDarknet(C2F)	800×800	11.2	44.7	27.1	58.4
YOLOv10-S [19]	CSPDarknet(SCDown)	800×800	7.2	33.7	26.4	57.2
YOLOv8-ADIL-S	**CSPDarknet(C2F, FIL)**	800×800	11.2	44.7	**28.4(+1.3)**	**60.2(+1.8)**
YOLOv10-ADIL-S	**CSPDarknet(SCDown, FIL)**	800×800	7.2	33.7	28.0(+1.6)	59.5(+2.3)
YOLOv8-M [17]	CSPDarknet(C2F)	800×800	25.9	123.3	29.7	62.6
YOLOv10-M [19]	CSPDarknet(SCDown)	800×800	15.4	92.4	28.9	61.6
YOLOv8-ADIL-M	**CSPDarknet(C2F, FIL)**	800×800	25.9	123.3	**31.9(+2.2)**	**64.9(+2.3)**
YOLOv10-ADIL-M	**CSPDarknet(SCDown, FIL)**	800×800	15.4	92.4	30.7(+1.8)	63.7(+2.1)
RetinaNet [56]	ResNet50	-	36.33	81.57	8.9	24.2
Faster R-CNN [26]	ResNet50	-	41.3	90.4	12.8	29.9
Cascade-RCNN [59]	ResNet50	-	62.15	121.86	15.1	34.2
DetecctoRS [61]	ResNet50	-	-	-	16.1	35.5
DotD [62]	ResNet50	-	-	-	20.4	51.4
NWD-RKA [12]	ResNet50	-	-	-	24.7	57.4
RFLA [50]	ResNet50	-	-	-	25.7	58.9
YOLOv8-L [17]	CSPDarknet(C2F)	800×800	43.7	258.1	29.8	63.7
DETR [52]	ResNet50	-	60.0	187.08	10.4	32.5
RT-DETR-RN50 [49]	ResNet50	640×640	42.0	130.5	18.5	49.7
Deformable-DETR [53]	ResNet50	-	40.0	173.0	18.9	50.0
DAB-DETR [63]	ResNet50	-	-	-	22.4	55.6
DINO-DETR [64]	ResNet50	-	-	-	25.9	61.3
DQ-DETR [65]	ResNet50	-	-	-	30.2	**68.6**
YOLOv10-L [19]	CSPDarknet(SCDown)	800×800	24.4	188.0	29.4	62.5
YOLOv8-ADIL-L	**CSPDarknet(C2F, FIL)**	800×800	43.7	258.1	**31.6(+1.8)**	65.9(+2.2)
YOLOv10-ADIL-L	**CSPDarknet(SCDown, FIL)**	800×800	24.4	188.0	30.9(+1.5)	64.3(+1.8)

remaining hyperparameters, including the temperature τ, the high-quality threshold t_{hq}, and the loss weight w_{IL}, together with the corresponding convergence behaviors, to demonstrate that ADIL provides stable improvements without requiring

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Fig. 2. Visualization results with VLM-imitation and without VLM-imitation.

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT EXEMPLAR GENERATION MECHANISMS

Exemplar Generation Mechanism	AP	AP_{50}	AP_{Small}	AP_{Medium}	AP_{Large}
Self-Imitation	25.5	42.8	15.7	38.9	46.0
VLM-Imitation	25.2	42.6	15.5	38.8	46.1
Self-Imitation+VLM-Imitation	**26.4**	**43.4**	**16.5**	**40.0**	**46.8**

delicate tuning. Unless otherwise specified, all experiments are conducted on the VisDrone2019-Det dataset using YOLOv10-ADIL-M.

1) Ablation on Dual-Exemplar Generation Mechanism: As shown in Table V, using self-imitation alone results in 25.5 AP, while VLM-imitation achieves 25.2 AP, indicating that both a data-adaptive exemplar bank and a semantic-stable prior are beneficial but insufficient by themselves. Notably, combining both methods—**Self-Imitation+VLM-Imitation**—achieves the best performance, reaching **26.4 AP** and **43.4** $\mathbf{AP_{50}}$, with the most obvious gain on small objects ($\mathbf{AP_{Small}}$: 15.7/15.5 $\rightarrow$ **16.5**), where HQ samples are typically sparse and exemplar updates are more unstable. This confirms the complementarity of the dual-exemplar design: self-imitation provides dataset-adaptive cues from reliable HQ samples, while VLM-imitation offers a stable semantic anchor that alleviates exemplar drift under cluttered remote sensing backgrounds. The visualization results in Fig. 2 further support this conclusion, showing that introducing VLM-imitation leads to more compact and category-consistent exemplars, whereas removing it results in noisier and less stable exemplar patterns, especially for small or partially occluded instances.

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT β'S

β	AP	AP_{50}	AP_{Small}	AP_{Medium}	AP_{Large}
0.5	26.1	42.8	16.0	39.7	46.3
1.0	26.2	42.7	15.9	39.8	46.2
2.0	**26.4**	43.2	16.4	40.0	**47.1**
4.0	26.3	43.3	16.3	39.9	46.5
6.0	**26.4**	**43.4**	**16.5**	40.0	46.8
8.0	26.1	43.1	16.4	**40.2**	46.4

2) Effect of β on Detection Performance: We evaluate the sensitivity of β in the quality score $t = f(s^{\alpha} \cdot u^{\beta})$, which controls how strongly HQ selection favors localization reliability (IoU). As shown in Table VI, ADIL is robust to a wide range of β (0.5–8.0), and the overall AP varies only marginally. More importantly, when β is relatively small (e.g., 0.5), HQ selection becomes more IoU-dominated, which tends to penalize samples with imperfect localization and may slightly hurt AP_{50} that is more sensitive to classification confidence under a fixed IoU threshold. In contrast, a larger β (e.g., 6.0) reduces the dominance of IoU in the quality score and allows high-confidence positives with reasonably good

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TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT LOSS FUNCTIONS

Loss	AP	AP_{50}	AP_{Small}	AP_{Medium}	AP_{Large}
$\mathcal{L}_{IIL}$	25.2	41.8	15.5	37.3	45.1
$\mathcal{L}_{CIL}$	25.6	42.5	15.7	37.4	46.0
$\mathcal{L}_{IIL} + \mathcal{L}_{CIL}$	**26.4**	**43.4**	**16.5**	**40.0**	**46.8**

TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT TEMPERATURE PARAMETERS

τ	AP	AP_{50}	AP_{Small}	AP_{Medium}	AP_{Large}
0.2	25.3	42.3	15.7	37.2	45.2
0.4	26.0	42.9	16.2	39.9	46.5
0.6	**26.4**	**43.4**	**16.5**	40.0	**46.8**
0.8	26.3	43.2	16.5	**40.2**	46.7
1.0	26.2	43.1	16.3	40.1	46.8

TABLE IX
IMPACT OF HIGH-QUALITY SAMPLE SELECTION THRESHOLD ON DETECTION PERFORMANCE

t_{hq}	AP	AP_{50}	AP_{Small}	AP_{Medium}	AP_{Large}
0.5	26.0	42.7	16.3	39.8	46.4
0.6	26.2	42.9	16.2	**40.2**	46.7
0.7	**26.4**	**43.4**	**16.5**	40.0	**46.8**
0.8	26.2	43.2	16.4	40.1	46.7
0.9	26.1	43.0	16.5	39.9	46.6

localization to be retained, leading to the best AP_{50} (43.4) and the highest AP_{Small} (16.5). Extremely large β (e.g., 8.0) may underemphasize localization cues, resulting in a mild drop. Overall, $\beta = 6.0$ achieves the best performance. Therefore, we adopt $\beta = 6$ as the default setting.

3) Effect of Different Loss Functions on Detection Performance: We perform ablation studies to assess the effectiveness of $\mathcal{L}_{IIL}$ and $\mathcal{L}_{CIL}$. As shown in Table VII, combining both losses achieves the best overall performance (AP: **26.4** and AP_{50}: **43.4**), with consistent improvements across small, medium, and large objects. This indicates that $\mathcal{L}_{IIL}$ and $\mathcal{L}_{CIL}$ provide complementary supervision. Instance-level matching improves sample-wise discrimination, while class-level alignment stabilizes intraclass consistency, leading to a more robust feature space.

4) Impact of the Temperature Parameter on Detection Performance: The temperature parameter $\boldsymbol{\tau}$ controls the sharpness of similarity distributions in contrastive learning. Table VIII shows that a moderate value of $\tau = 0.6$ yields the best balance between AP (**26.4**) and AP_{50} (**43.4**). Importantly, the performance remains relatively stable when $\tau \in [0.6, 1.0]$ (AP varies within 0.2), suggesting low sensitivity around the optimum. In contrast, smaller τ (e.g., 0.2) produces a sharper distribution that tends to overemphasize hard positives/negatives, which may lead to noisier gradients and slower or less stable convergence; overly large τ smooths the distribution and weakens discrimination, resulting in marginally degraded performance. Therefore, we set $\tau = 0.6$ by default as a robust choice.

5) Impact of High-Quality Sample Selection Threshold on Robustness: The threshold $\boldsymbol{t_{hq}}$ determines which instance features are considered high-quality for updating class exemplars. As shown in Table IX, the best overall result is achieved at $t_{hq} = 0.7$ (AP: **26.4** and AP_{50}: **43.4**). We observe a mild plateau for $t_{hq} \in [0.6, 0.8]$, indicating that the method is not overly sensitive to this parameter. When t_{hq} is too low, lower quality samples are included, which may introduce noise into exemplar updates and reduce final accuracy; when t_{hq} is too high, the number of selected samples decreases, increasing optimization variance and slightly harming medium-scale performance (e.g., $t_{hq} = 0.9$). Based on the tradeoff between filtering noise and maintaining sufficient sample diversity, we use $t_{hq} = 0.7$ as the default.

TABLE X
IMPACT OF LOSS WEIGHT ON IL

w_{IL}	AP	AP_{50}	AP_{Small}	AP_{Medium}	AP_{Large}
0.25	25.5	42.7	15.8	37.6	45.7
0.5	26.4	**43.4**	**16.5**	40.0	46.8
0.75	**26.5**	43.2	16.4	40.0	**46.9**
1.0	26.0	43.0	16.1	**40.1**	46.4

6) Impact of IL Loss Weight and Optimization Tradeoff: The loss weight $\boldsymbol{w_{IL}}$ adjusts the influence of the IL objective during optimization. Table X shows that extremely small weights (e.g., 0.25) underutilize the IL supervision and lead to noticeably lower AP, while overly large weights may over-regularize the detector and slightly reduce AP_{50}. Notably, $w_{IL} = 0.75$ achieves the highest AP (**26.5**), whereas $w_{IL} = 0.5$ yields the best AP_{50} (**43.4**) and also provides the strongest AP_{Small} (**16.5**). Since our goal is to improve overall detection quality while preserving strong localization performance at IoU = 0.5 and small-object accuracy, we select $w_{IL} = 0.5$ as the default setting, which offers a better balanced tradeoff and stable convergence in practice.

E. Qualitative Study

Beyond quantitative improvements in metrics such as AP and AP_{50}, our method also demonstrates clear advantages in perceptual quality. To provide an intuitive understanding of the improvements brought by our proposed feature imitation learning, we visualize the detection results of the baseline YOLOv10-M and the enhanced YOLOv10-ADIL-M on representative examples from the VisDrone2019-Det dataset. As shown in Fig. 3, YOLOv10-ADIL-M exhibits a substantially lower false detection rate in the first column, where YOLOv10-M produces multiple spurious boxes. In the second and third columns, YOLOv10-ADIL-M significantly outperforms the baseline in detecting weak and densely distributed targets, benefiting from improved feature representation and stronger discriminative capabilities. The results indicate notable reductions in both false positives and false negatives.

To further validate the model's effectiveness in challenging scenarios, we present the output heatmaps using the official code in Ultralytics [66] in Fig. 4. Compared with YOLOv10-M, YOLOv10-ADIL-M captures a broader set of relevant regions, especially for small-scale objects that are otherwise missed by the baseline model. These visualizations clearly highlight the effectiveness of the proposed method in enhancing feature focus and improving detection robustness in complex scenes.

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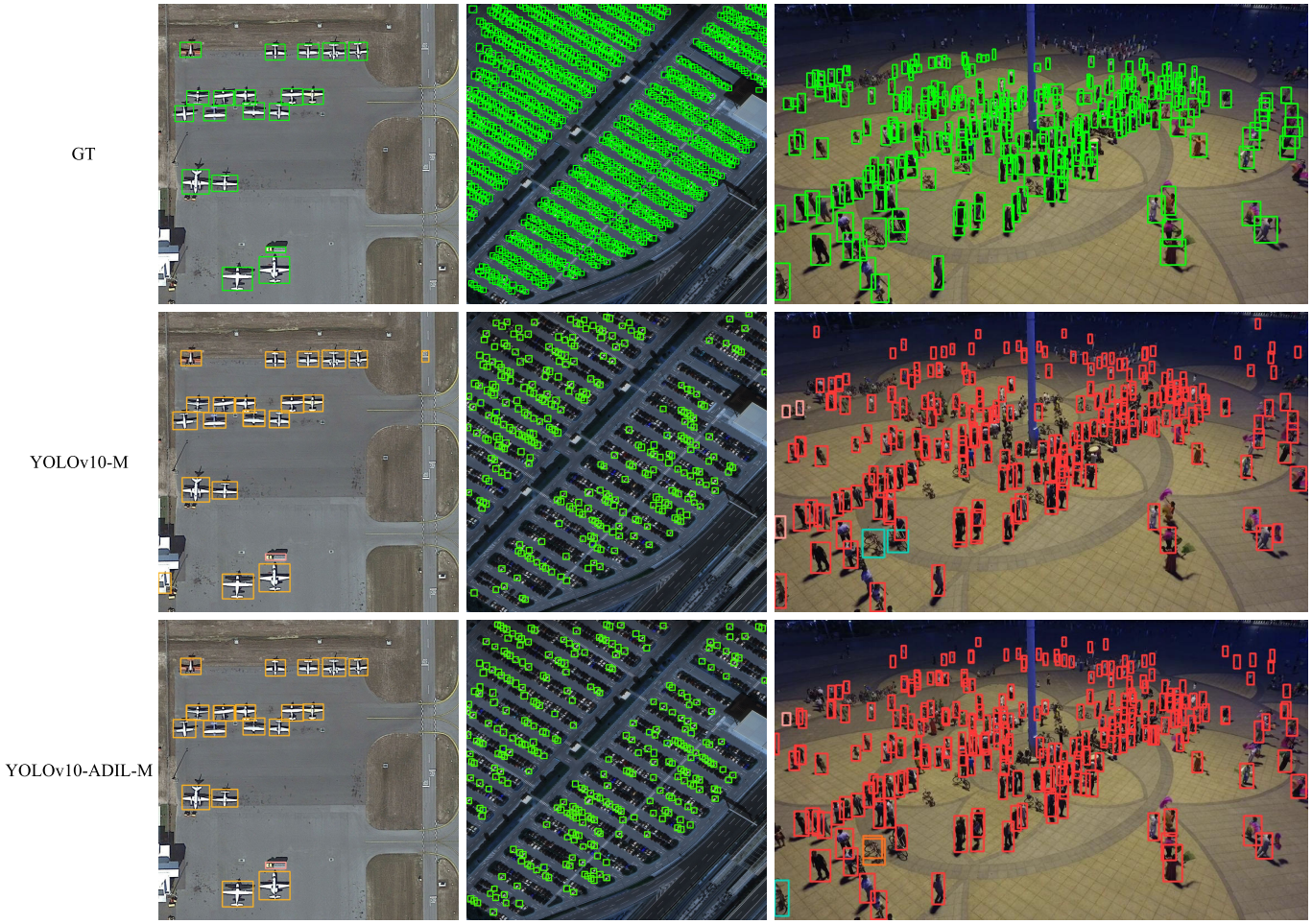


Fig. 3. YOLOv10-ADIL-M delivers more accurate detection than YOLOv10-M, particularly in dense or low-contrast scenarios.

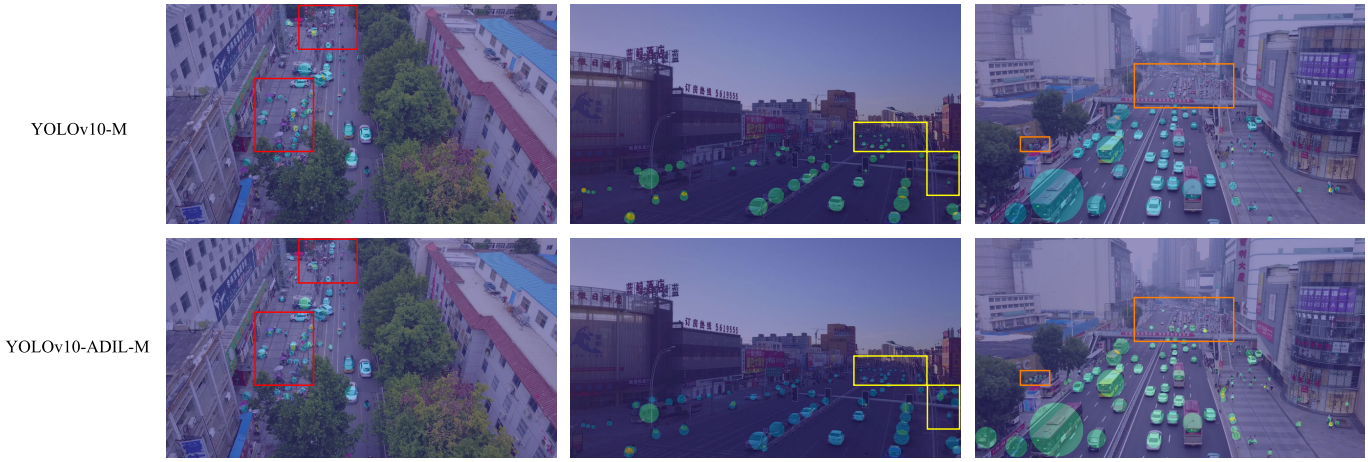


Fig. 4. YOLOv10-ADIL-M yields clearer attention maps and better detects small, dense targets on VisDrone2019-Det.

V. Conclusion

This article proposed an ADIL strategy to enhance the detection performance of weak and small targets in remote sensing scenarios. ADIL integrated self-imitation and VLM-imitation, leveraging high-quality sample features extracted by the network and the textual space of VLM to construct discriminative exemplars that guide network learning. When integrated into YOLO-v8/YOLO-v10, the strategy achieved significant performance improvements across various network architectures and multiple remote sensing datasets. Moreover, our method introduced no additional parameters or

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computational costs during testing, maintaining high inference speed and strong practical application value.

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